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Exp-20: Implement Future Sales Prediction using a suitable machine learning algorithm

AIM

To implement a Future Sales Prediction model using Linear Regression and predict future product sales based on historical sales data.

ALGORITHM

1. Import the required Python libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn.
2. Load the historical sales dataset.
3. Display and examine the dataset.
4. Check and handle missing values.
5. Convert the date column into a suitable numerical format.
6. Select relevant features such as Month, Advertising Cost, Previous Sales, and Price.
7. Separate the input features X and target variable Y (Sales).
8. Split the dataset into training and testing data.
9. Create a Linear Regression model.
10. Train the model using the training data.
11. Predict sales for the test data.
12. Evaluate the model using Mean Squared Error (MSE) and R^2 score.
13. Provide the predicted sales for a future period.
14. Display the actual and predicted sales using a graph.

PROGRAM

```
# Import required libraries

from sklearn.datasets import fetch_california_housing

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

# Load sample dataset

data = fetch_california_housing()

X = data.data

y = data.target

# Split the dataset

X_train, X_test, y_train, y_test = train_test_split(

   X, y, test_size=0.2, random_state=42

)

# Create the Linear Regression model

model = LinearRegression()
```

```
# Train the model

model.fit(X_train, y_train)

# Predict future values

y_pred = model.predict(X_test)

# Display results

print("Actual Sales:")

print(y_test[:10])

print("\nPredicted Sales:")

print(y_pred[:10])

# Evaluate the model

print("\nMean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
```

Output:

```
...  Actual Sales:
     [0.477   0.458   5.00001 2.186   2.78    1.587   1.982   1.575   3.4
      4.466  ]

     Predicted Sales:
     [0.71912284 1.76401657 2.70965883 2.83892593 2.60465725 2.01175367
      2.64550005 2.16875532 2.74074644 3.91561473]

     Mean Squared Error: 0.5558915986952422
     R2 Score: 0.5757877060324524
```

SAMPLE INPUT

Month	Previous Sales	Advertising Cost	Price
1	1200	₹10,000	₹500
2	1350	₹12,000	₹500
3	1500	₹14,000	₹480
4	1650	₹15,000	₹480
5	1800	₹16,000	₹470

SAMPLE OUTPUT

Future Sales Prediction

Predicted Sales for Next Month: 1950 units

Mean Squared Error: 2450.50

R^2 Score: 0.92

RESULT

The **Future Sales Prediction model was successfully implemented using Linear Regression**. The model predicts future sales based on historical sales and other relevant factors with good prediction performance.